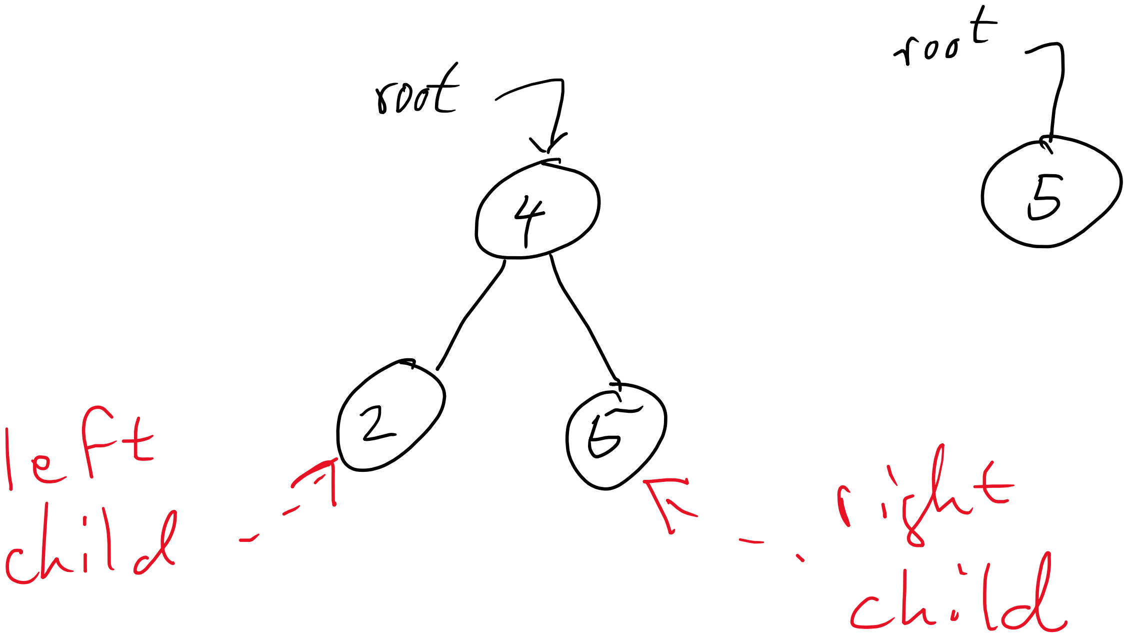
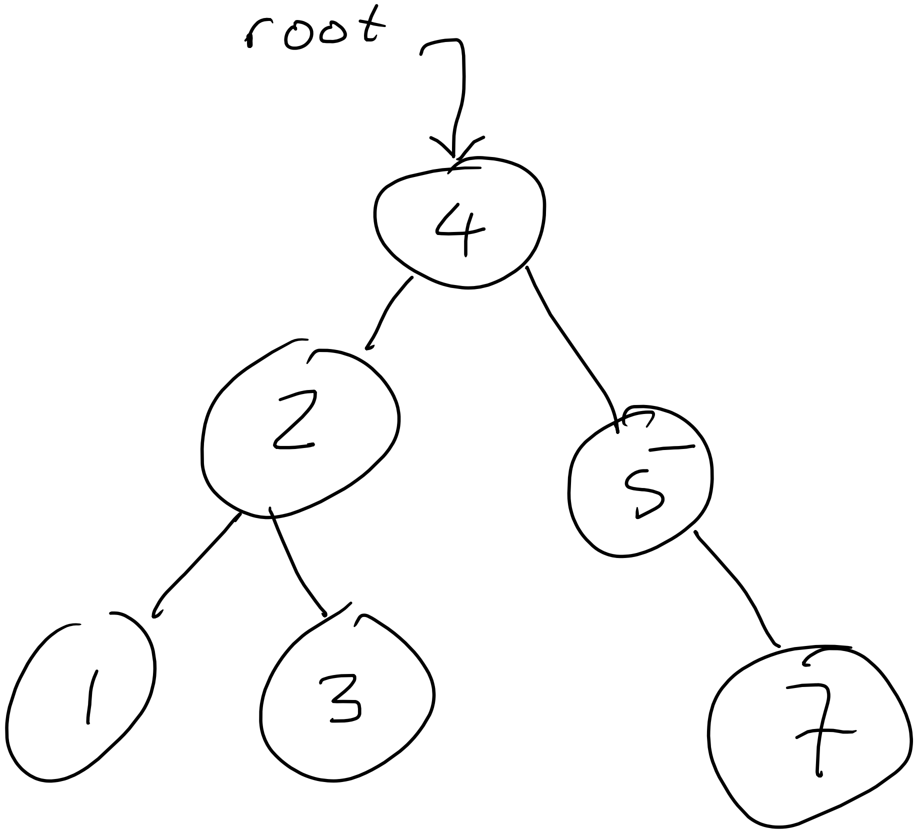
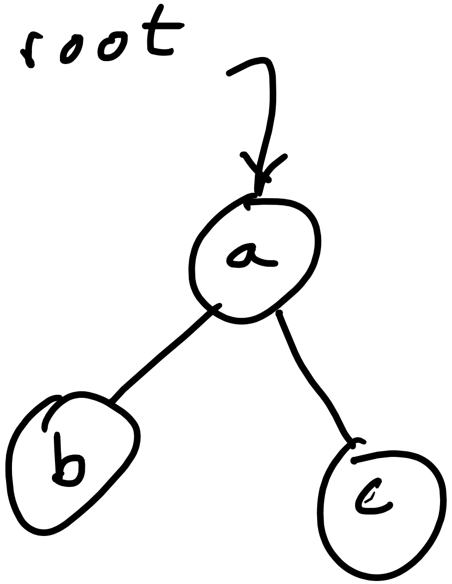
BST – Binary Search Tree







0, 1 or 2 children per node

$b < a$ and $c > a$

Why are trees great?

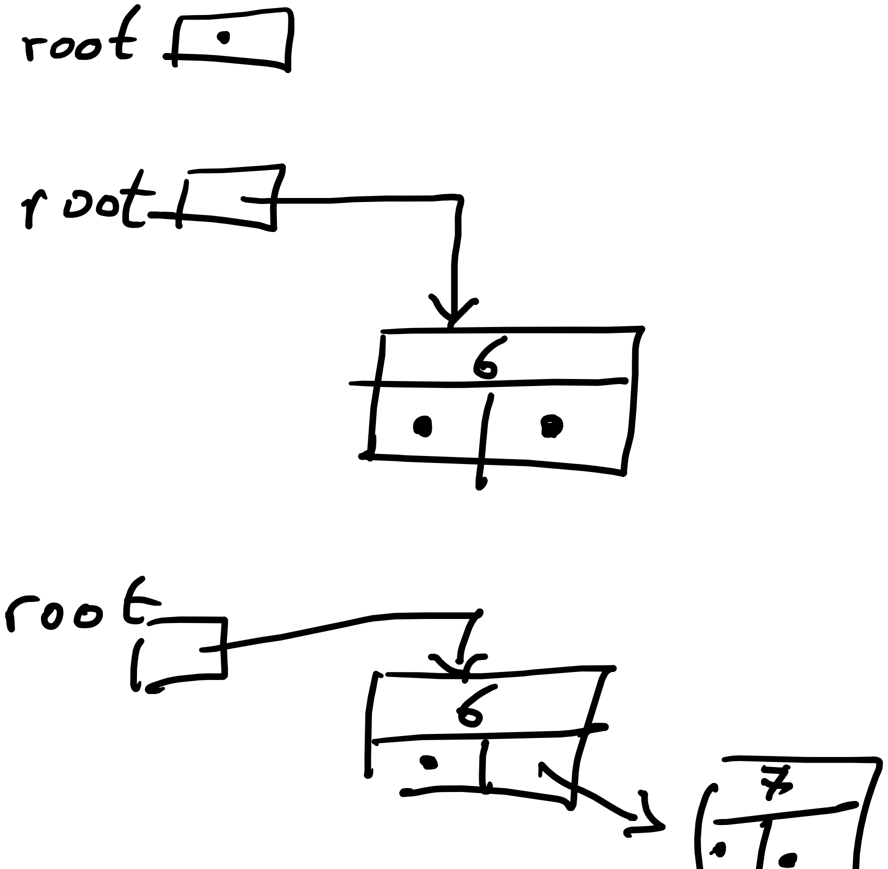
- Searchable in O(log n) time.

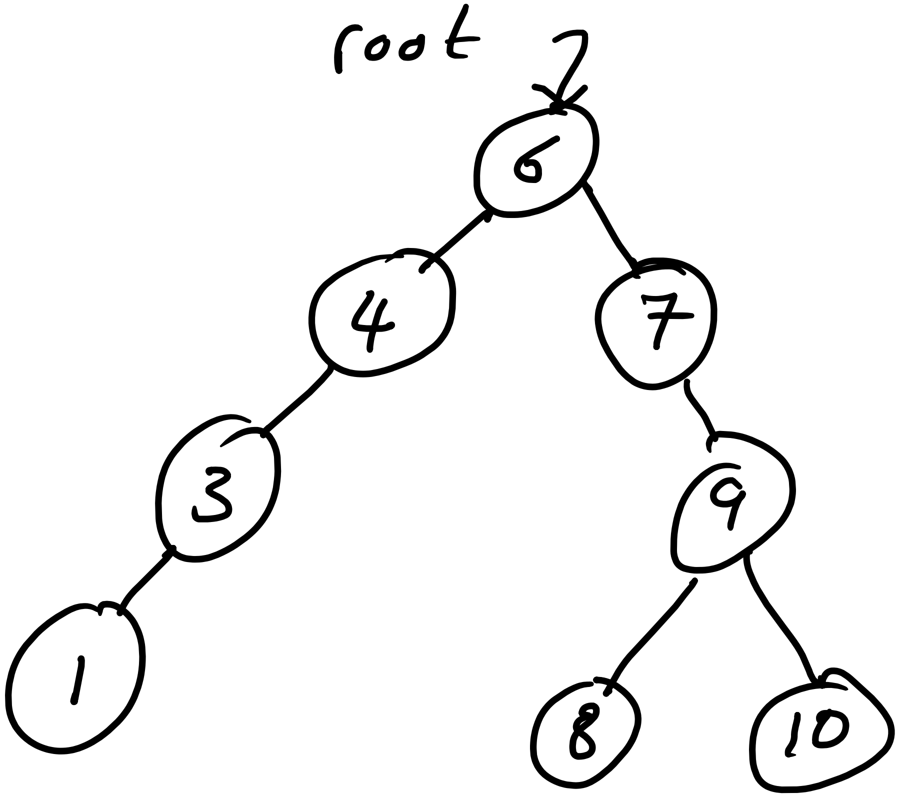
BST operations (methods)

- Insert an element
- Delete or remove an element
- Search or find an element
- Peek at the root
- Traversals
  - Inorder
  - Preorder
  - Postorder
- Depth First Search
- Breadth First Search

Insert operation

- Insert the following numbers into a tree of integers
- 6, 7, 4, 9, 10, 3, 1, 8



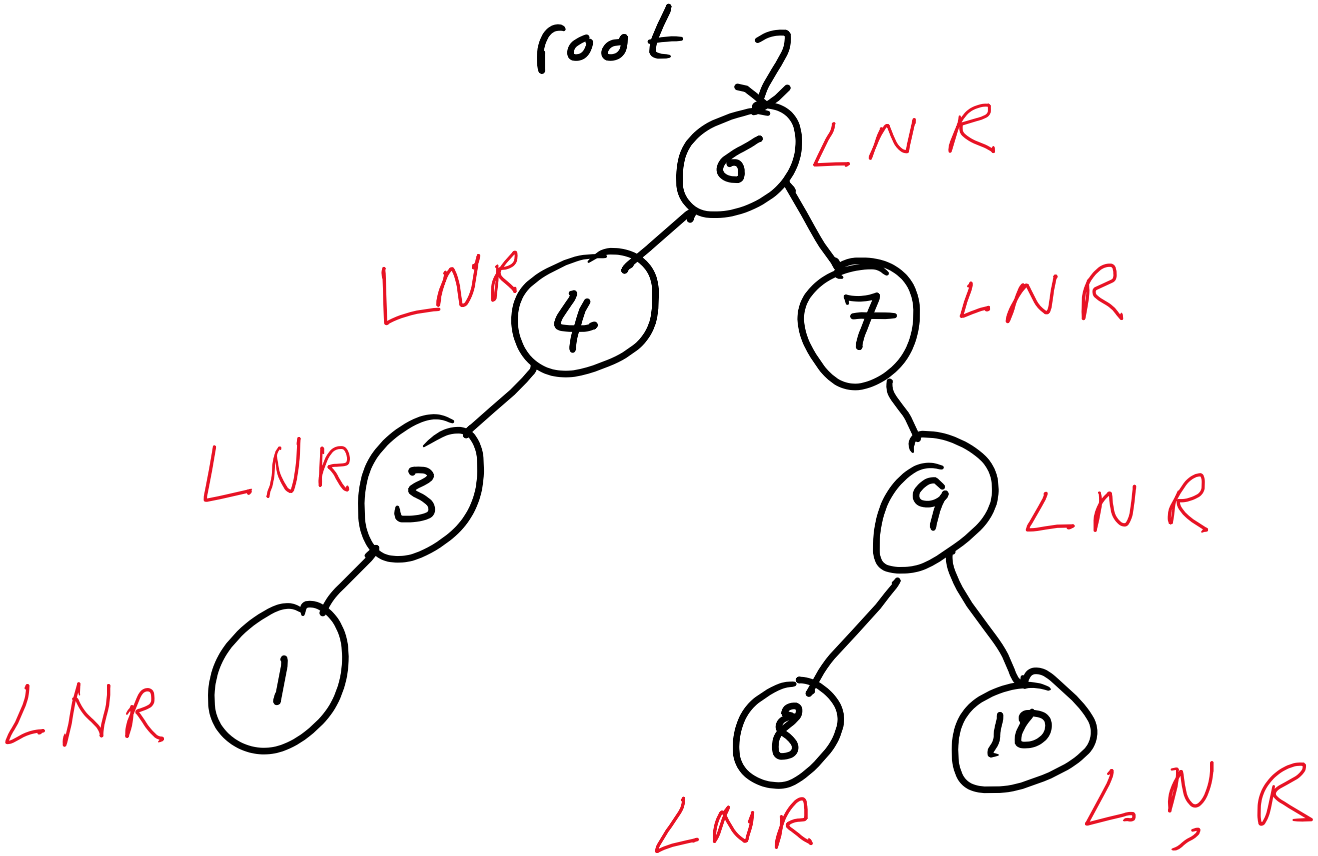


Inorder Traversal or Visitation is a Recursive Procedure.

L 1. Go Left

N 2. Visit Node (Print)

R 3. Go Right

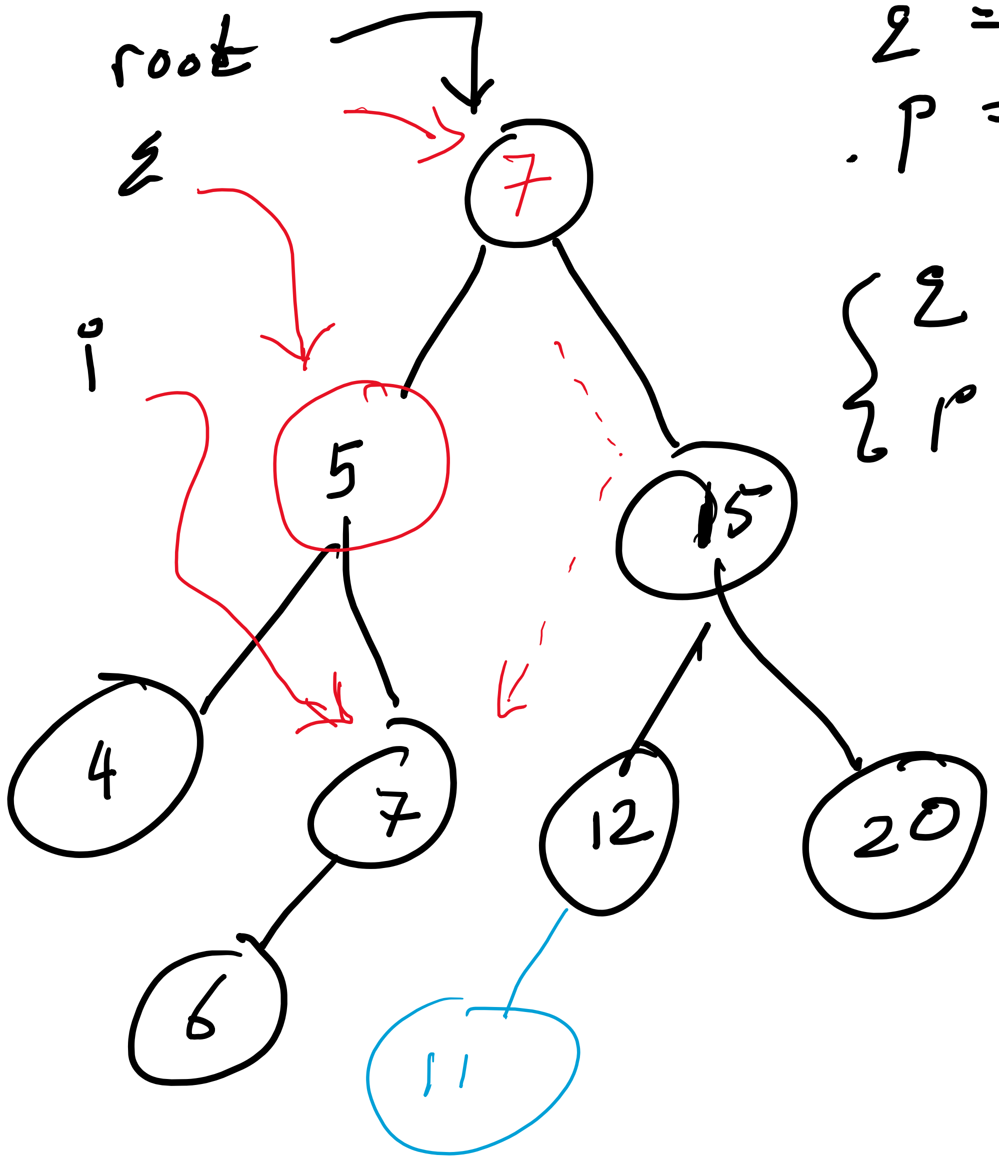


Print Output: 1, 3, 4, 6, 7, 8, 9, 10

Preorder NLR

PostOrder LRN

Inorder LNR



q = null

p = root

$\begin{cases} q = p \\ p = p.getLeft(); \end{cases}$

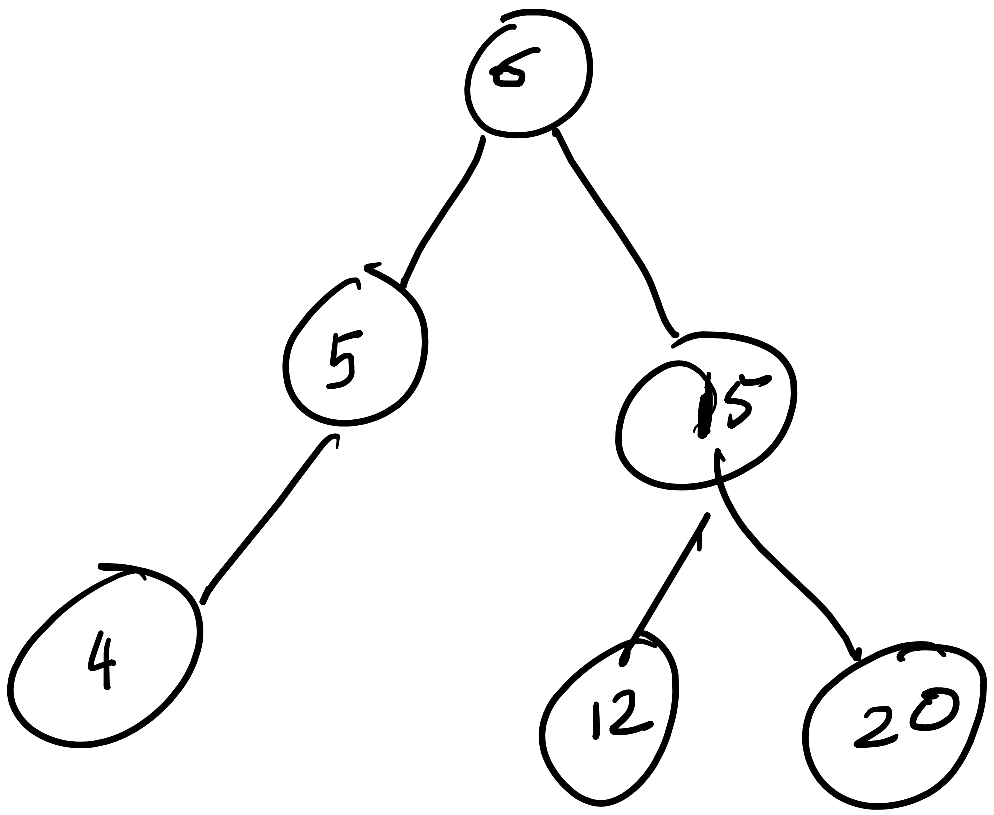
q = p;

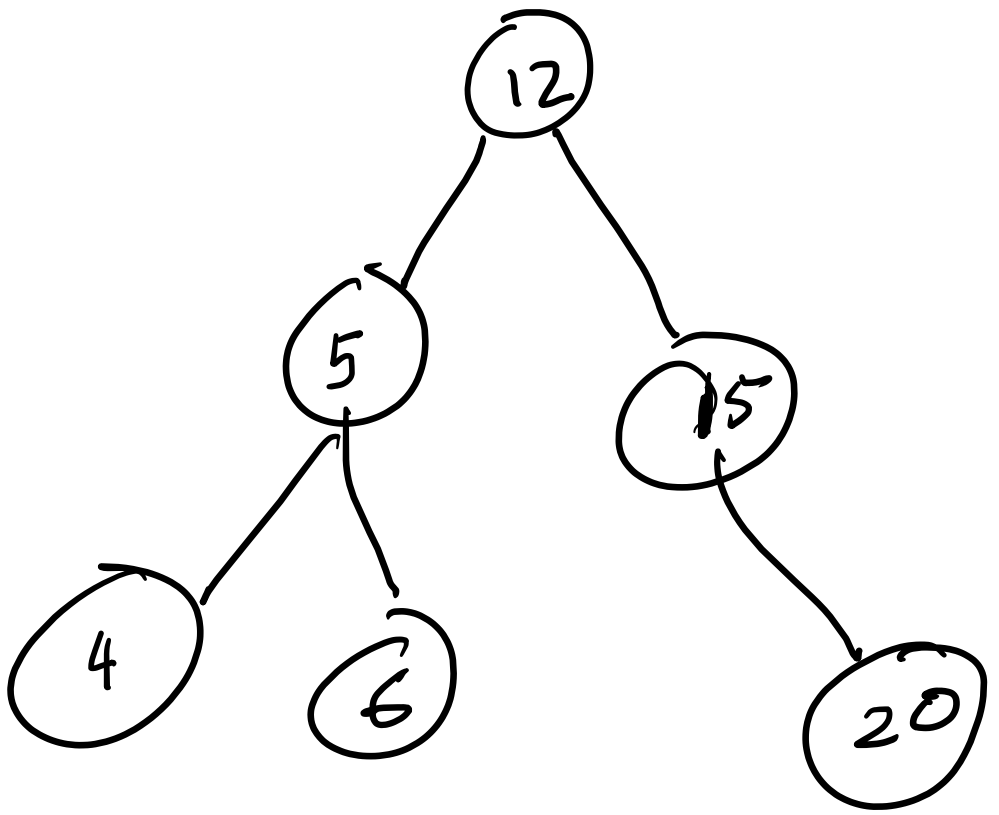
p = p.getRight()

root.getValue(p.getValue());

q.setRight(p.getLeft());

deleting 10





delete 10